



THE UNIVERSITY OF
MELBOURNE

Marketing Research

Week 6



Recap

	Face-to-face	Web surveys
Costs	☐	☒
Max. Length	☒	☐
Interviewer Bias	☐	☒
Sample Bias	☐ - ☒	☐
Open Answers	☒	☐
Complexity (Filter)	☐	☒
Flexibility of methods	☐	☒

Qualtrics

Researchers view: ☐ ☐ ☐ ☒ Very bad Very good

1. Screeners or classification questions come first. These questions determine which parts of the survey a respondent should fill out.

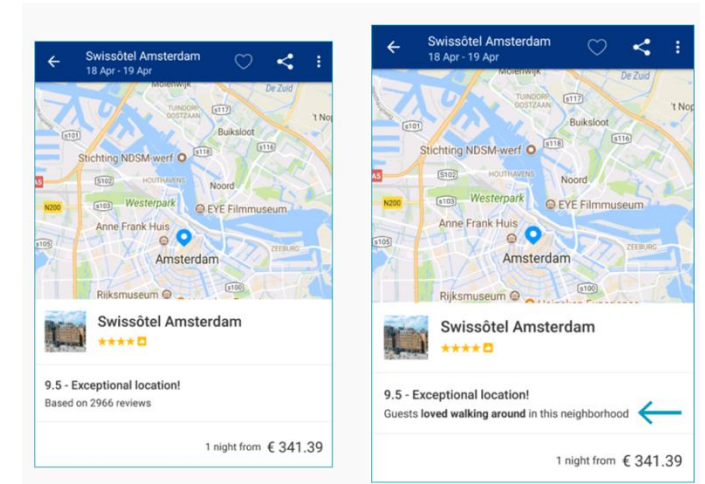
2. Next, insert the key variables of the study. This includes the dependent variables, followed by the independent variables.

3. Use the funnel approach. That is, ask questions that are more general first and then move on to details. This makes answering the questions easier as the order aids recall. Make sure that sensitive questions are put at the very end of this section.

4. Demographics are placed last if they are not part of the screening questions.

A: The Control
Shows the site's current practice

B: The Treatment
Adds walkability information



Before-after experiment
with a control group:

Experimental group (R)	O_1	X	O_2
Control group (R)	O_3		O_4



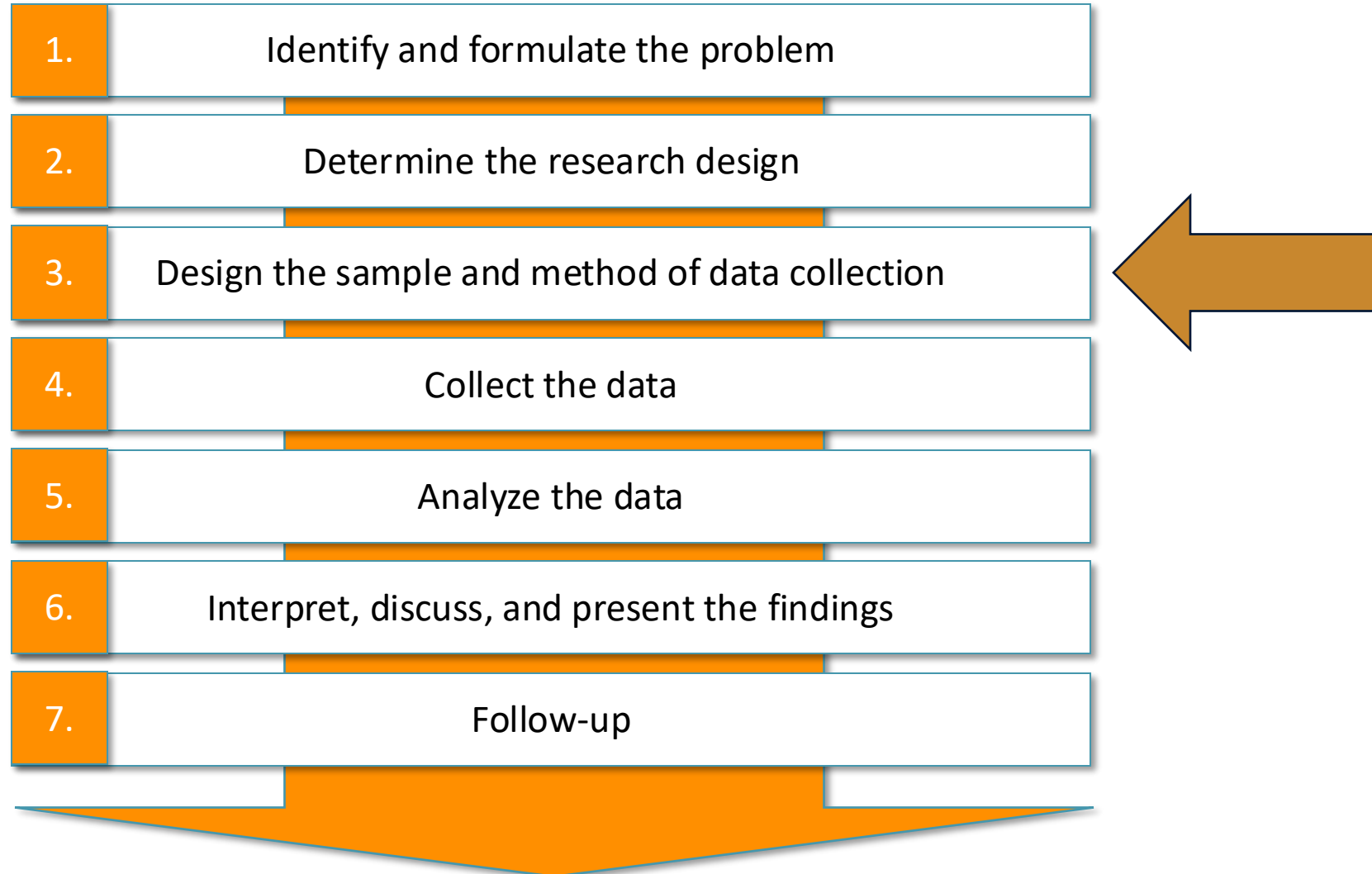
Topics for Today

1. Determining sampling methods
2. Determining sample sizes
3. Avoiding nonresponse
4. Workshop in SPSS
5. Group work



Determining sampling methods

The Market Research Process



Sampling

Population is the group of units about which we want to make judgments

Sampling is the process through which we select cases from a population

- Why do we draw units from a population?
- How do we draw units from a population?



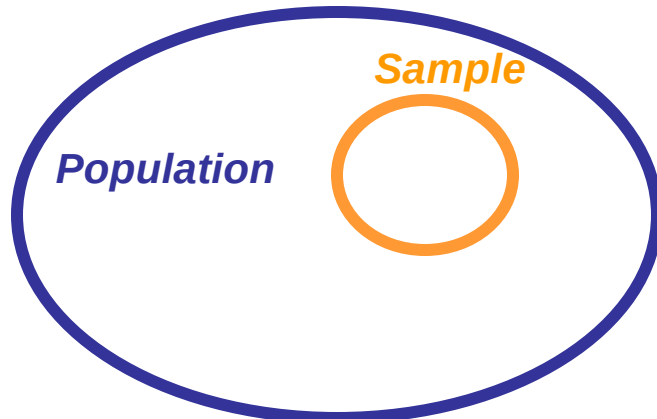
Population versus sample

Population: The entire group of individuals in which we are interested.

Example: All humans, all working-age people in Melbourne, all dog owners

Census: data collected from every member of the population

A **parameter** is a number describing a characteristic of the **p**opulation (e.g., population mean)



Sample: The part of the population we actually examine and for which we do have data.

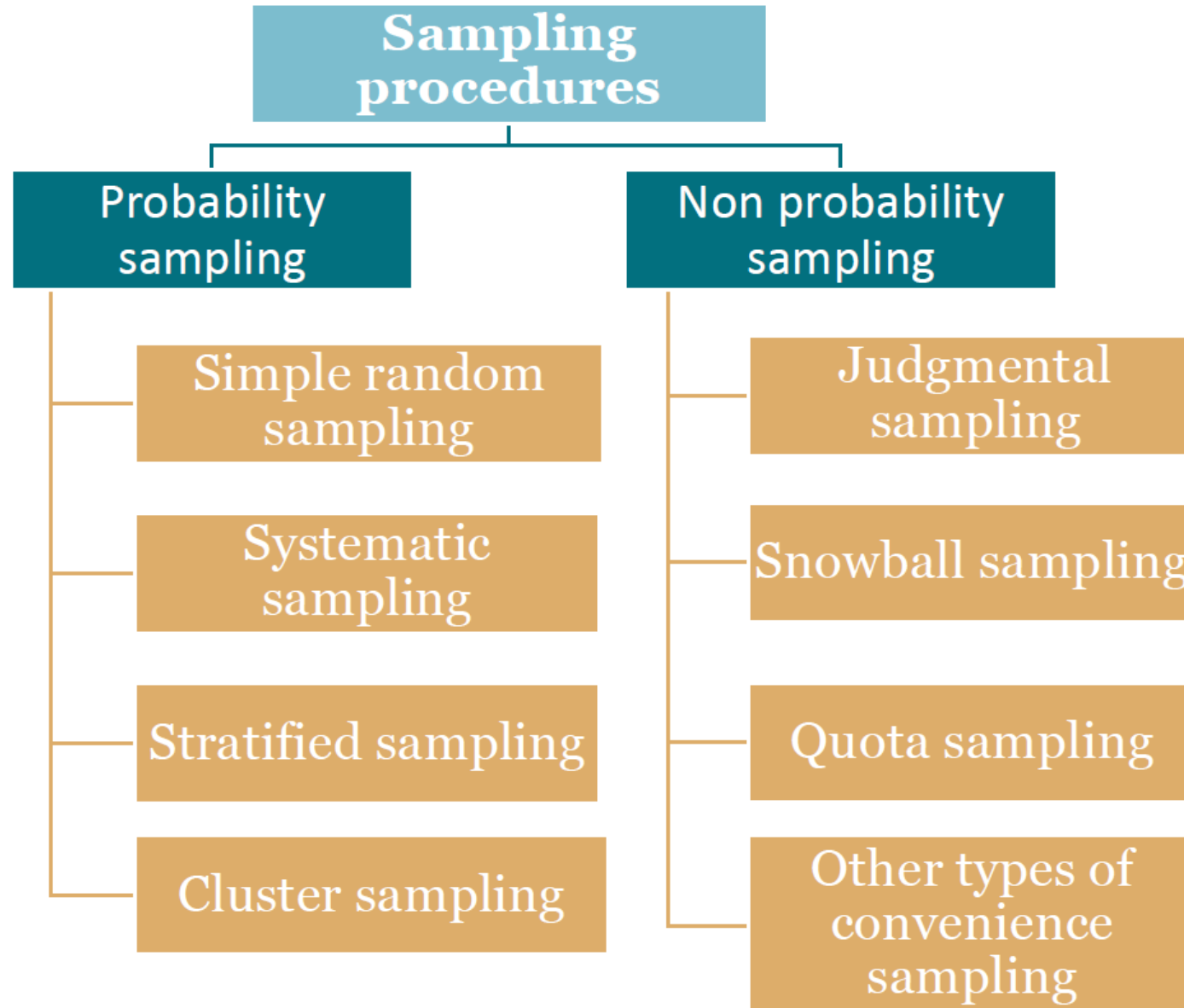
A **statistic** is a number describing a characteristic of a **s**ample (e.g., sample mean)

How well the sample represents the population depends on the **sample design**.

Types of Sampling

- **Probability sampling**
 - Every element of the known population has a known, nonzero probability of being selected.
 - Sample is representative. Thus, findings are projectable to the entire population
 - Sampling error can be computed, and inferential statistics can be calculated
- **Nonprobability sampling**
 - Cannot make inferences about the general population
 - Generally used for qualitative research
 - convenience, judgment, quota, snowball

Sampling Procedures





Sampling frame

Sampling frame: List of every object in the target population

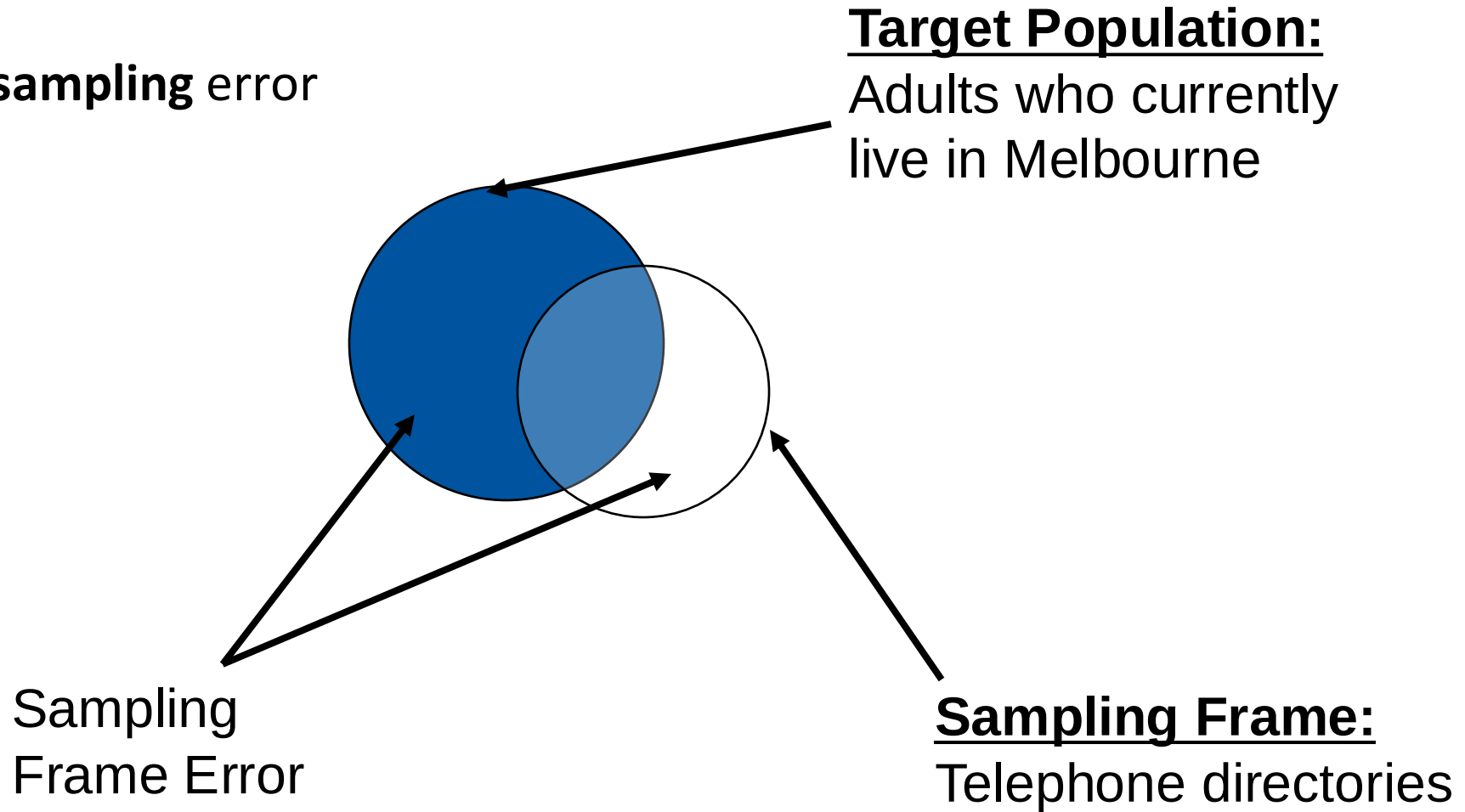
- Can be expensive and time-consuming, but very important

How to obtain a list of every object?

- E.g., membership list (for existing members), voter list (for election)
- It is difficult to be exhaustive
 - Rare events may not make the list – e.g., drug users, undocumented immigrants

Sampling Frame Error

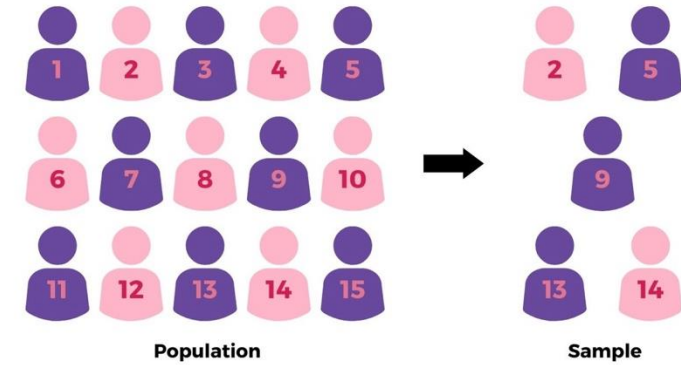
A type of **non-sampling** error



Simple Random Sampling

- ❖ Purest form of random sampling
- ❖ Selected by rolling dice, roulette wheel, drawing names from a hat, random number generation
- ❖ Need a list naming and identifying the population (where to find these?)
- ❖ Seldom used because of inconvenience

Simple Random Sampling



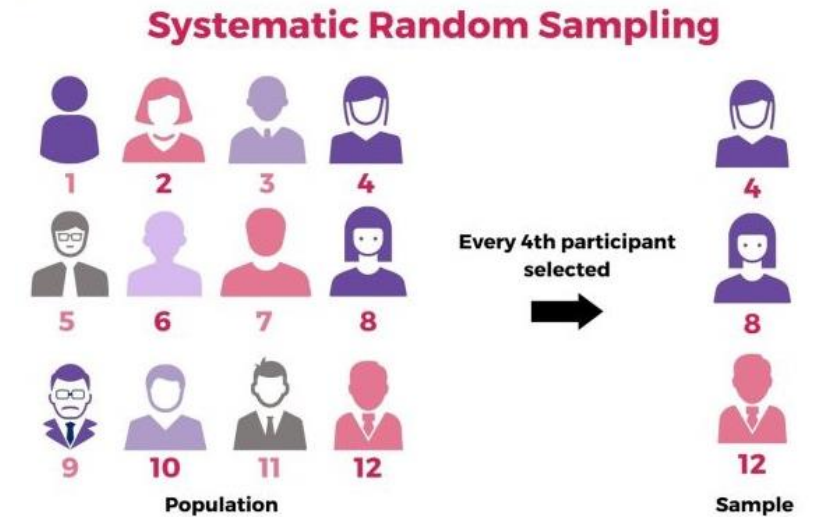


Simple Random Sampling

- ❖ Each element in the population has an equal chance to be selected.
- ❖ **Advantage:** Good generalizability and unbiased results
- ❖ **Disadvantage:** difficult to obtain a complete, current and accurate listing of the target population elements (i.e., sampling frame). Each sampling unit needs to be identified.
- ❖ Application: random-digit-dialing (RDD) technique to generate random telephone numbers.

Systematic Sampling

- ❖ Substitute for, and more practical than simple random sampling
- ❖ n^{th} name selection
- ❖ Skip interval = population size divided by sample size
- ❖ When to use?
 - Restricted budget
 - Results needed in a short timeframe
 - Overall population is large → hard to obtain sampling frame
 - Low risk of data manipulation



Skip Interval = 4

Systematic Sampling

❖ Advantages

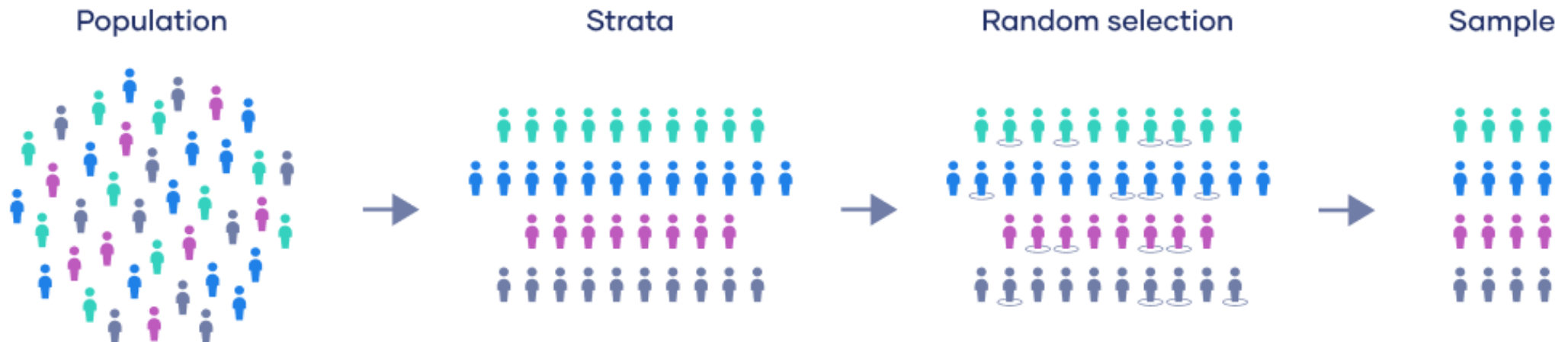
- Simple and quick implementation
- Samples are evenly distributed
- Less opportunity for data manipulation

❖ Disadvantages

- Size of the population cannot be determined accurately
- Natural degree of randomness needed.
 - Example: if odd number = male; even number = female. What is the potential problem?

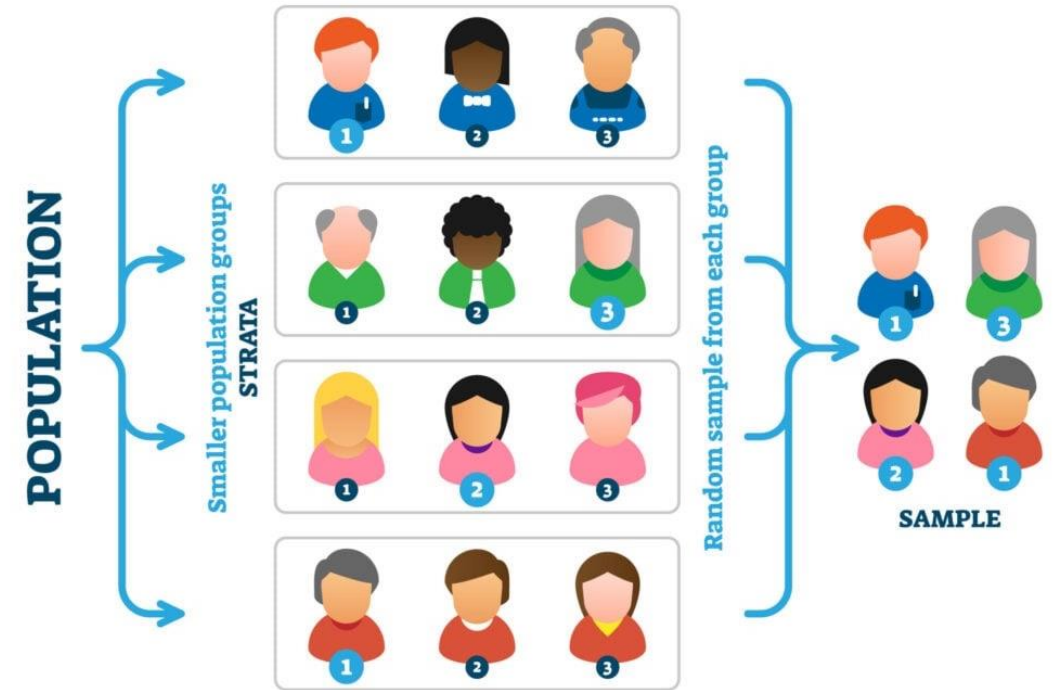
Stratified Sampling

- ❖ Reduce sampling error by splitting the population into groups
- ❖ Process:
 - ✓ Identify classification factor
 - ✓ Split population by classification factor
 - ✓ Draw random (or systematic) samples from each group
- ❖ Results in **homogeneous-within, heterogeneous between** sample groups



Stratified Sampling

- ❖ Advantage:
 - ✓ Ensure important strata are in the sample
 - ✓ The sample is more representative (greater precision)
- ❖ Disadvantage:
 - ✓ Sometimes challenging to find the basis for stratifying
 - ✓ Implementation can be hard if there are too many strata
- ❖ Example:
 - ✓ For new products, divide the potential consumers based on age/gender etc.

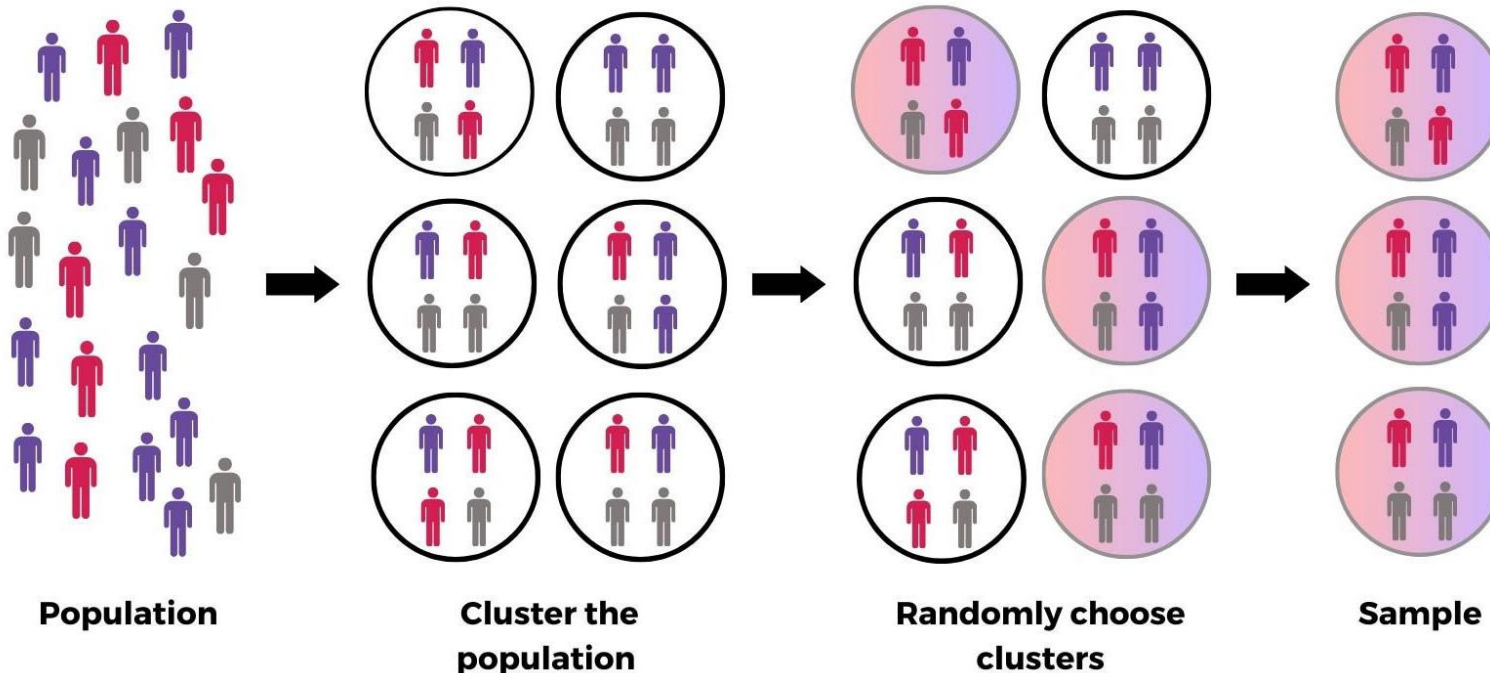


Cluster Sampling

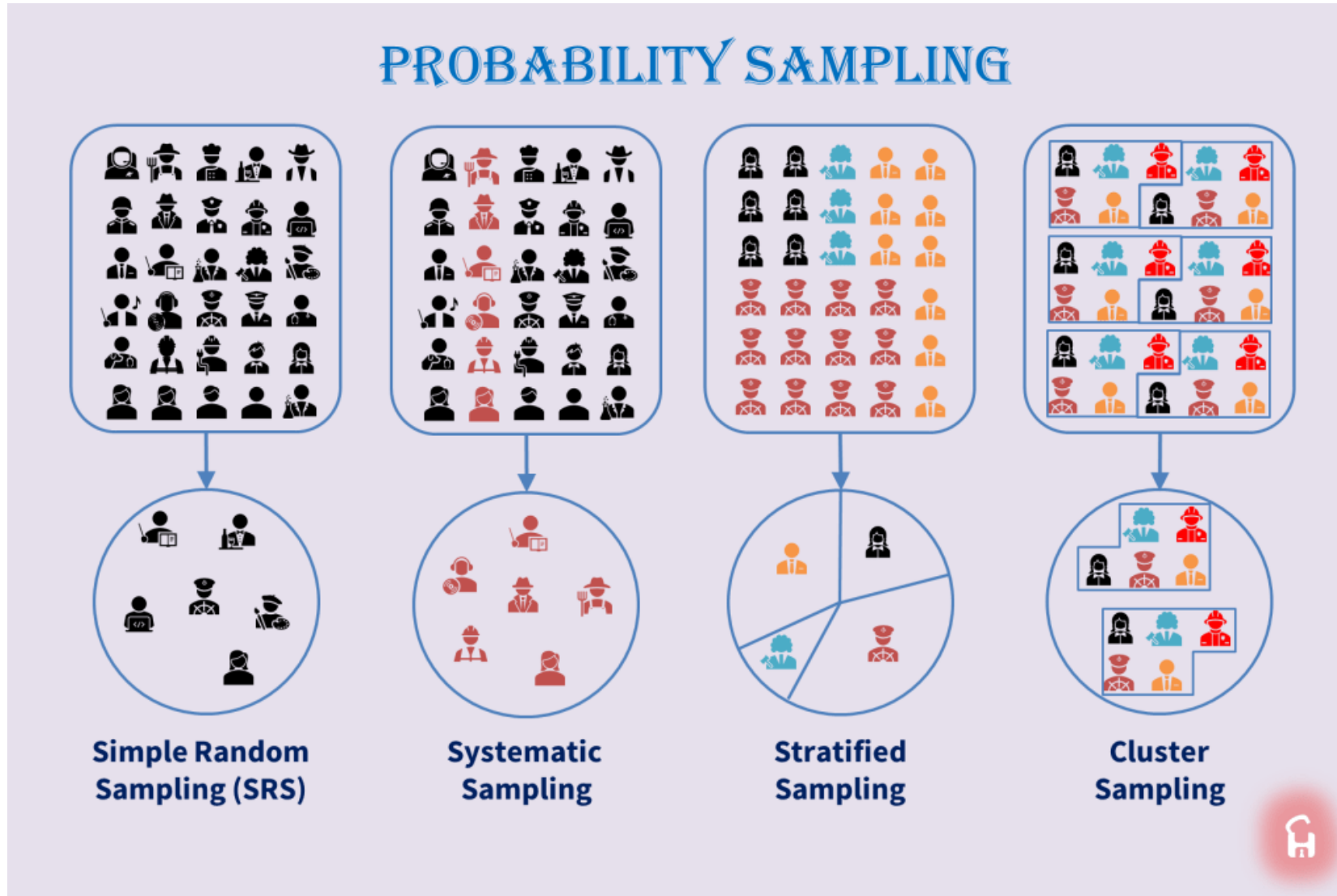
❖ Process:

- ✓ Identify representative clusters of the population (e.g., counties, cities, city blocks)
- ✓ Draw a random sample from one or several clusters (instead of every cluster)

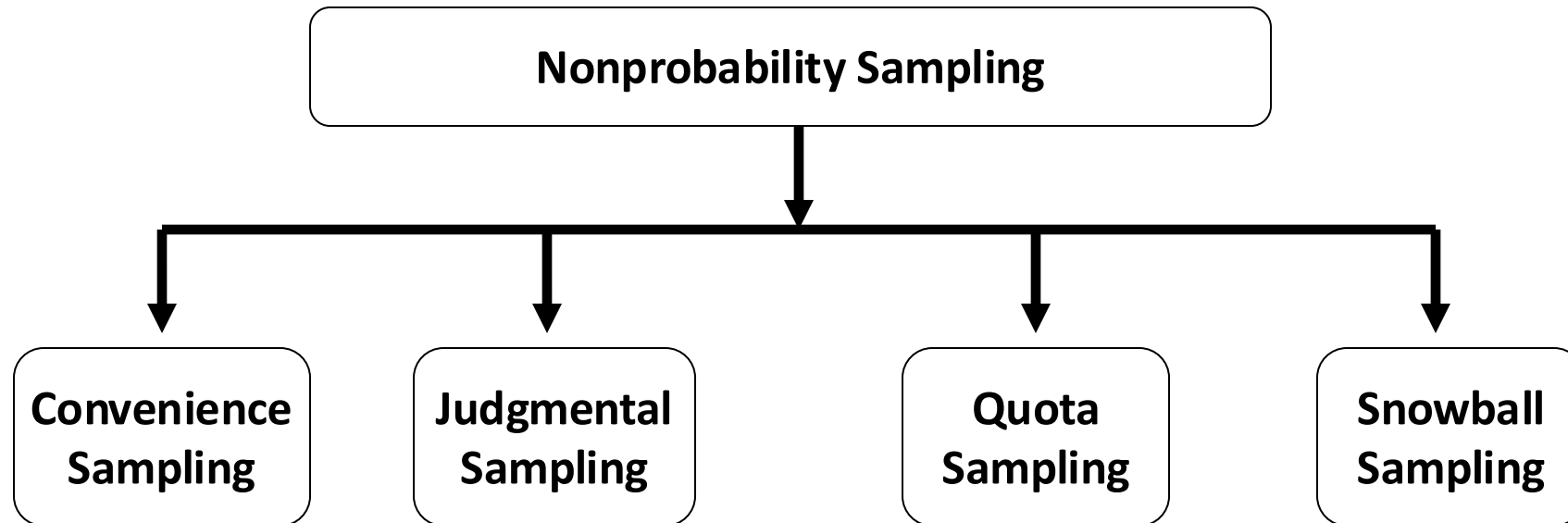
❖ Creates **heterogeneous-within, homogeneous between** sample groups



Sampling Type Comparison



Nonprobability Sampling Techniques



NO guarantee that your sample will be representative.

Whether an element enters the sample or not depends on researcher's judgement.

Convenience Sampling (Accidental)

A sample of convenient elements

- They happen to be in the right place at the right time
- e.g., the students in the front row, “people-on-the-street” interview, mall intercept interview

Advantages

- Least expensive
- Least time consuming

Disadvantages

- Not representative → hard to generalize findings





Judgmental Sampling (purposive)

The elements/subjects are selected based on the judgment of the researcher

- Examples: test markets, expert witness in the court, subjects in the **focus group**
- Popular in qualitative studies

Advantages

- Low cost and still less time consuming than most of the techniques

Disadvantages

- Results depend on the judgment of the researcher
- No guarantee for a representative sample



Quota Sampling

Two-step process

- Develop quotas for each control characteristic
- Convenience or judgment sampling to fulfill the quota

In short, it is like a “restricted” judgmental sampling

- Pros and cons are like the ones in judgmental sampling

Example:

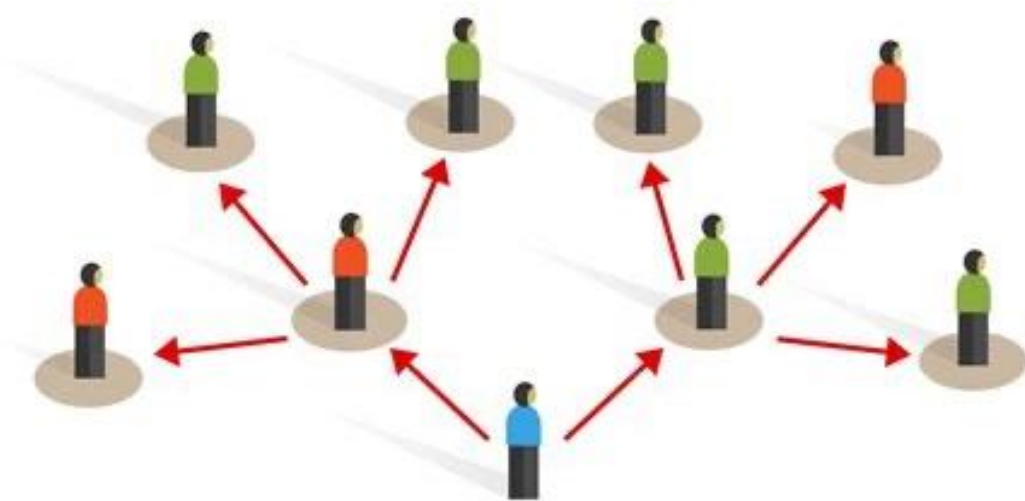
Control Characteristic: Gender	Percentage in population	Quota Sample
Male	42%	420
Female	58%	580
	100%	1,000

Snowball Sampling (referral sampling)

Select an initial group

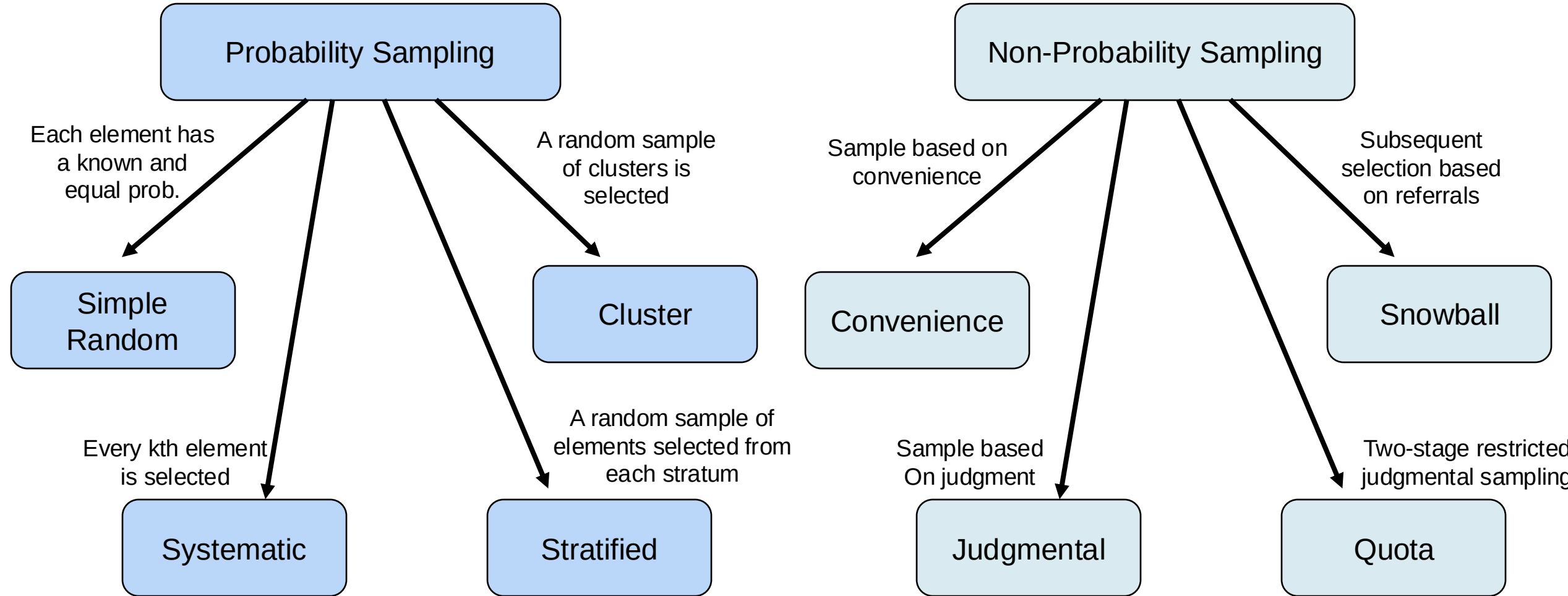
The subsequent subjects are selected based on the referrals

Example:

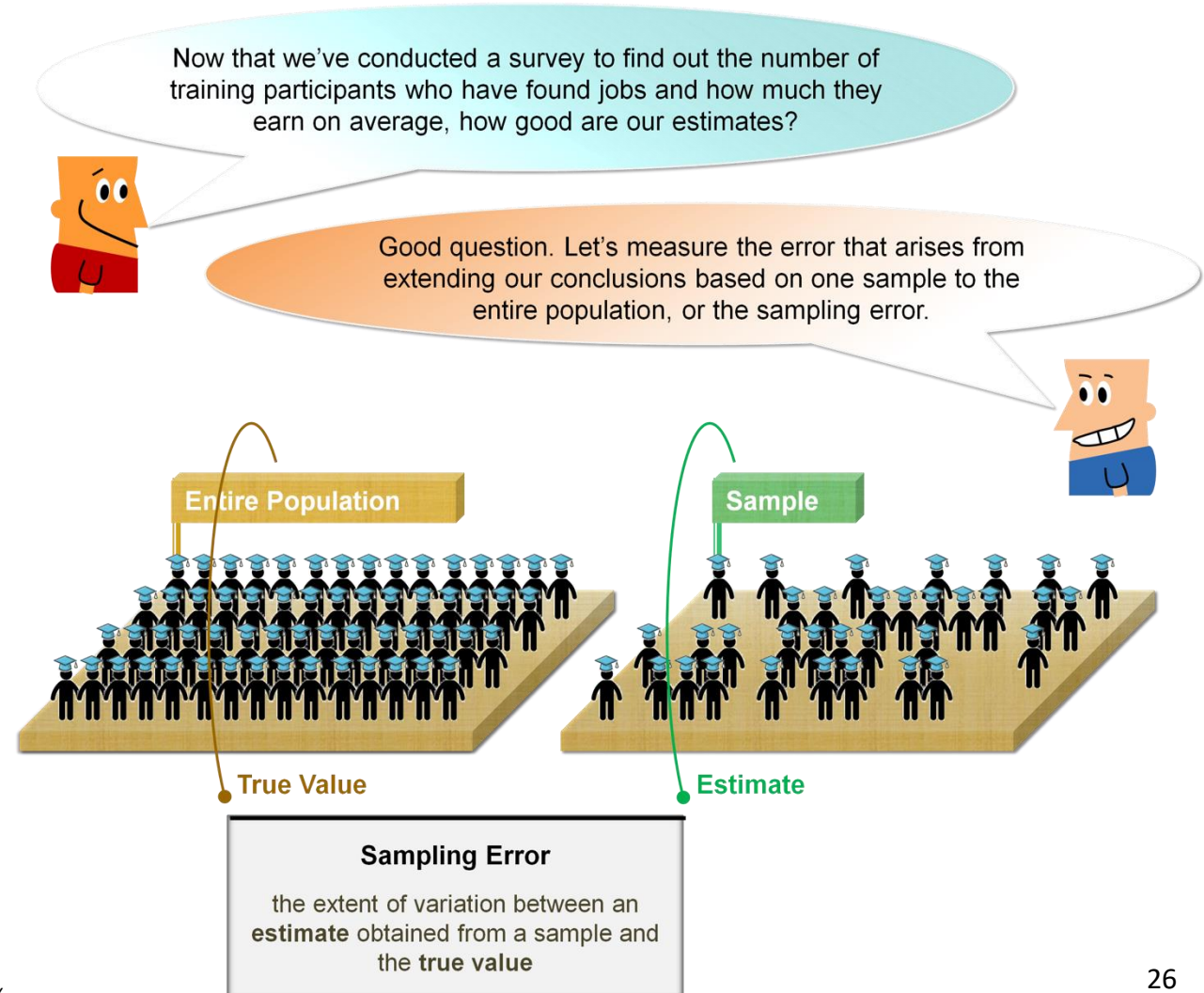
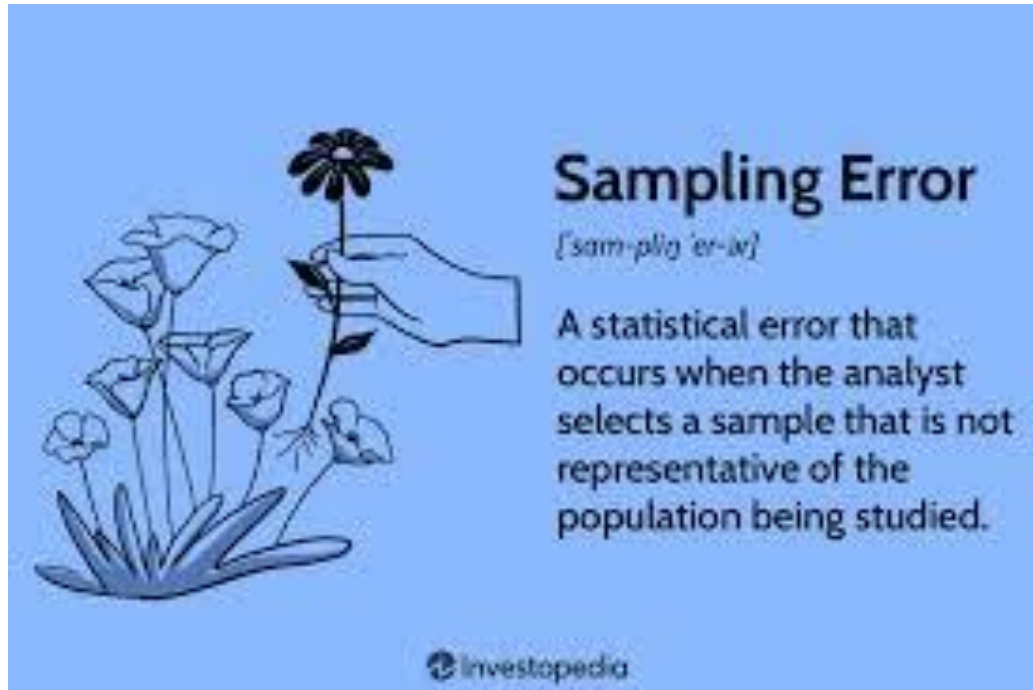


- Advantages
 - Hard-to-reach population
 - (difficult to obtain sampling frame)
 - e.g., drug users, people with double nationality, undocumented immigrants, field experts
- Disadvantages
 - Difficult to obtain the initial sample
 - Community bias → no guarantee for representative sample

Summary



Sampling Error





Sampling Error (calculation)

How accurate is the sample result?

What is the percentage of people who have ever reviewed a restaurant? (probability)

- Sample1 = 70%
- Population = 73%
- Sampling error= 3%

How many times have you reviewed a restaurant in the last 3 months? (mean)

- Sample1 = 5.8
- Population = 5.5
- Sampling error= 0.3

sources to conduct a poll. But calling an election wrongly – as when YouGov forecast in 2004 that John Kerry would beat George Bush by 3 per cent – brings the wrong sort of publicity. “We were at the edge of sampling error,” Kellner says. “It was not our finest hour. The Republican right was just more effective than we realised in getting its vote out.”

Embarrassments like that make it vital to the business that they read the views of the voters correctly, and that depends on finding an accurate sample. As



Determining sample sizes

Sample Sizes

- ❖ Intuition: Increasing the sample size of a randomly selected sample helps reduce sampling error.
- ❖ The marginal benefit of increasing the sample sizes becomes smaller as the sample becomes larger. An increase from $n=40$ to $n=60$ (i.e., + 50%) is more beneficial than from $n=1,000$ to $n=1,020$ (i.e., + 2%).
- ❖ The sample size required is (almost) unrelated to the size of the population.

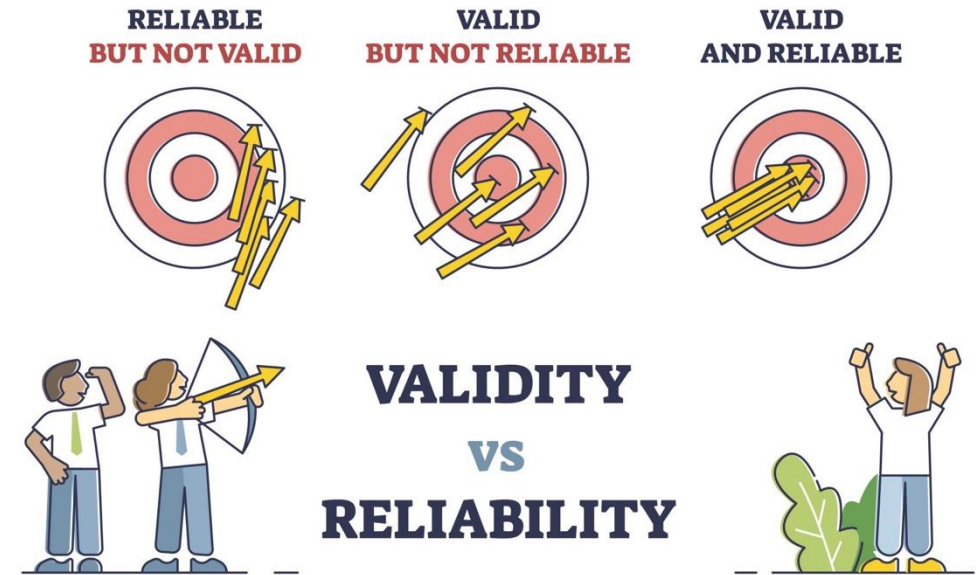


Determining Acceptable Sample Sizes (1)

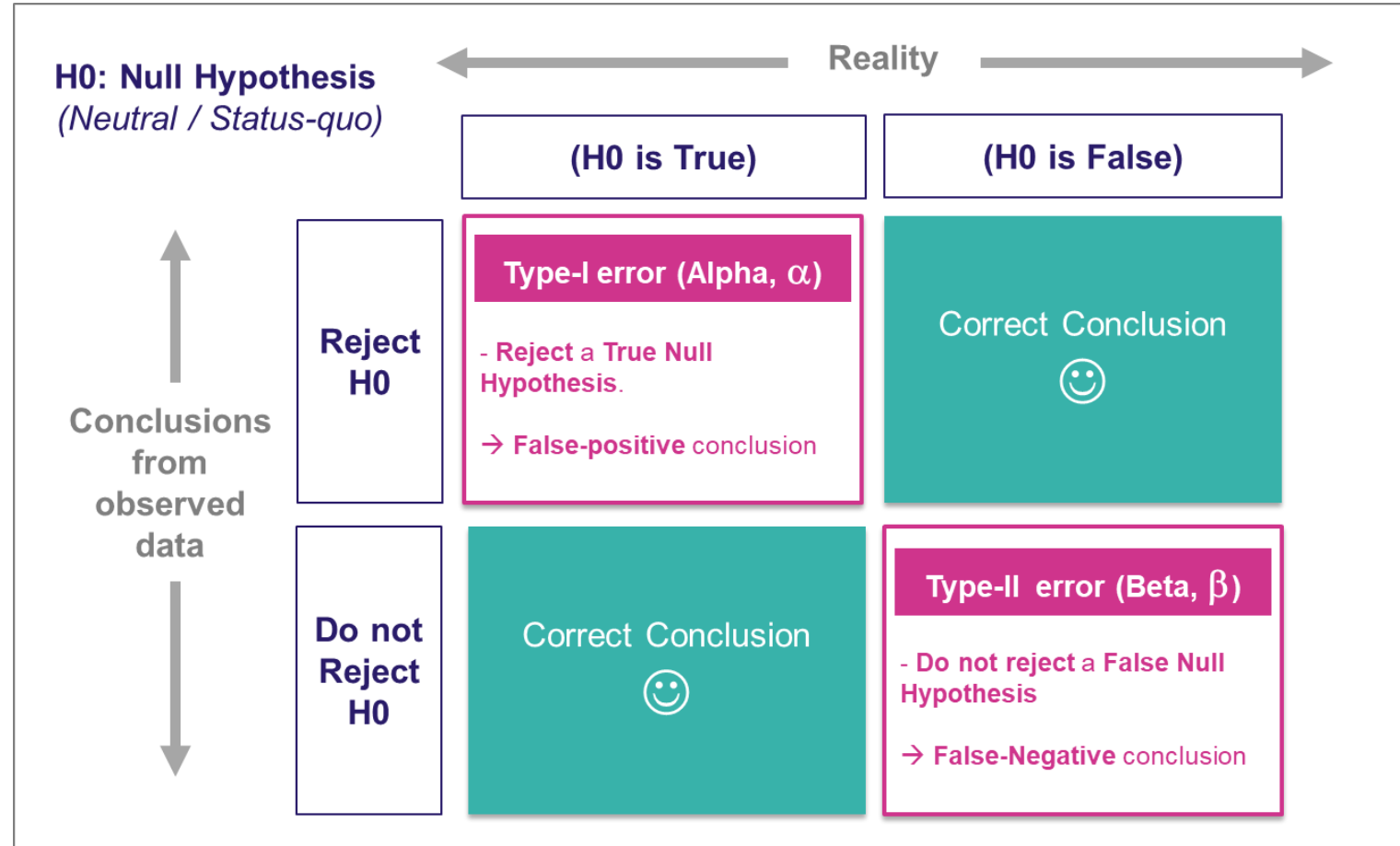
Collecting more data provides greater reliability but does not improve validity.

However, the cost of greater samples is high.

- Web surveys \$0,35-\$10,00 per additional respondent
- Face-to-face \$50 -\$200 per additional respondent
- Larger samples take longer time to collect – may not support quick decision-making
- Could hurt validity in extreme cases



Principles of Hypothesis Testing



Determining Acceptable Sample Sizes (2)

- ❖ Calculate the expected nonresponse
 - ✓ For example: we need 100 usable responses. If we expect the response to be 10%, approach 1000 potential respondents.
- ❖ Sample size recommendations are per group. If you want to compare 2 groups, take twice the sample size.
- ❖ Be aware of bots participating. Surveys on social media or websites are likely to attract these. (see Paas, L. J., & Morren, M. (2018). Please do not answer if you are reading this: Respondent attention in online panels. *Marketing Letters*, 29, 13-21.)



Calculate CI and Sample Sizes

Q1. Have you ever left an online review for a restaurant that you visited?

___ Yes ___ No

Q2. How many times in the past 3 months have you left a review for a restaurant?

___ times

What kind of scale type?

Margin of Error (E) and Confidence Interval (CI)

Margin of Error (E) = maximum expected difference between the true population value and the sample estimate — due to random sampling.

The margin of error is influenced by

- ✓ Confidence level (e.g., z-value)
- ✓ Population standard deviation (σ)
- ✓ Sample size (n)

Confidence Interval (CI) = The range within which the true population value likely falls.

$$CI = \text{Point Estimate} \pm \text{Margin of Error}$$

Confidence Interval (mean)

Q2. *How many times in the past 3 months have you left a review for a restaurant?* _____
times

$$\text{CI for mean} = \bar{x} \pm z \frac{\sigma}{\sqrt{n}}$$

- To be 90% confident, z value = 1.65
- To be 95% confident, z value = 1.96
- To be 99% confident, z value = 2.58

Assume

- Sample=100
- Sample mean=5.8
- Sample standard deviation=1.2

$$\begin{aligned}\text{Confidence interval (95\%)} &= 5.8 \pm 1.96 * [1.2 / \text{sqrt}(100)] \\ &= 5.8 \pm 1.96 * 0.12 \\ &= 5.8 \pm 0.24 \\ &= (5.56, 6.04)\end{aligned}$$

Use Confidence Interval (mean) to Calculate Sample Size

- If the margin of error is specified, we can use the confidence interval formula to calculate the required sample size.

$$E = z \frac{\sigma}{\sqrt{n}} \quad \longrightarrow \quad n = \left(\frac{(z)(\sigma)}{E} \right)^2$$

- Example: Suppose the standard deviation (σ) is known to be 5. What is the sample size required to construct a 95% confidence interval with a margin of error (E) of 1?
- $n = (1.96 \times 5 / 1)^2 = 96.04 \rightarrow 97$

Confidence Interval (proportion)

Q1. Have you ever reviewed a restaurant that you visited? ____Yes ____ No

$$\text{Confidence interval for a proportion} = \hat{p} \pm z \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Assume:

- Sample size (n) = 200
- Sample proportion ($\hat{p}$) = 76%

$$\begin{aligned}\text{Confidence interval (95\%)} &= 76\% \pm 1.96 * \text{sqrt}[(76\% * 24\%) / (200)] \\ &= 76\% \pm 5.90\% \\ &= (70.1\%, 81.9\%)\end{aligned}$$

Use Confidence Interval (proportion) to Calculate Sample Size

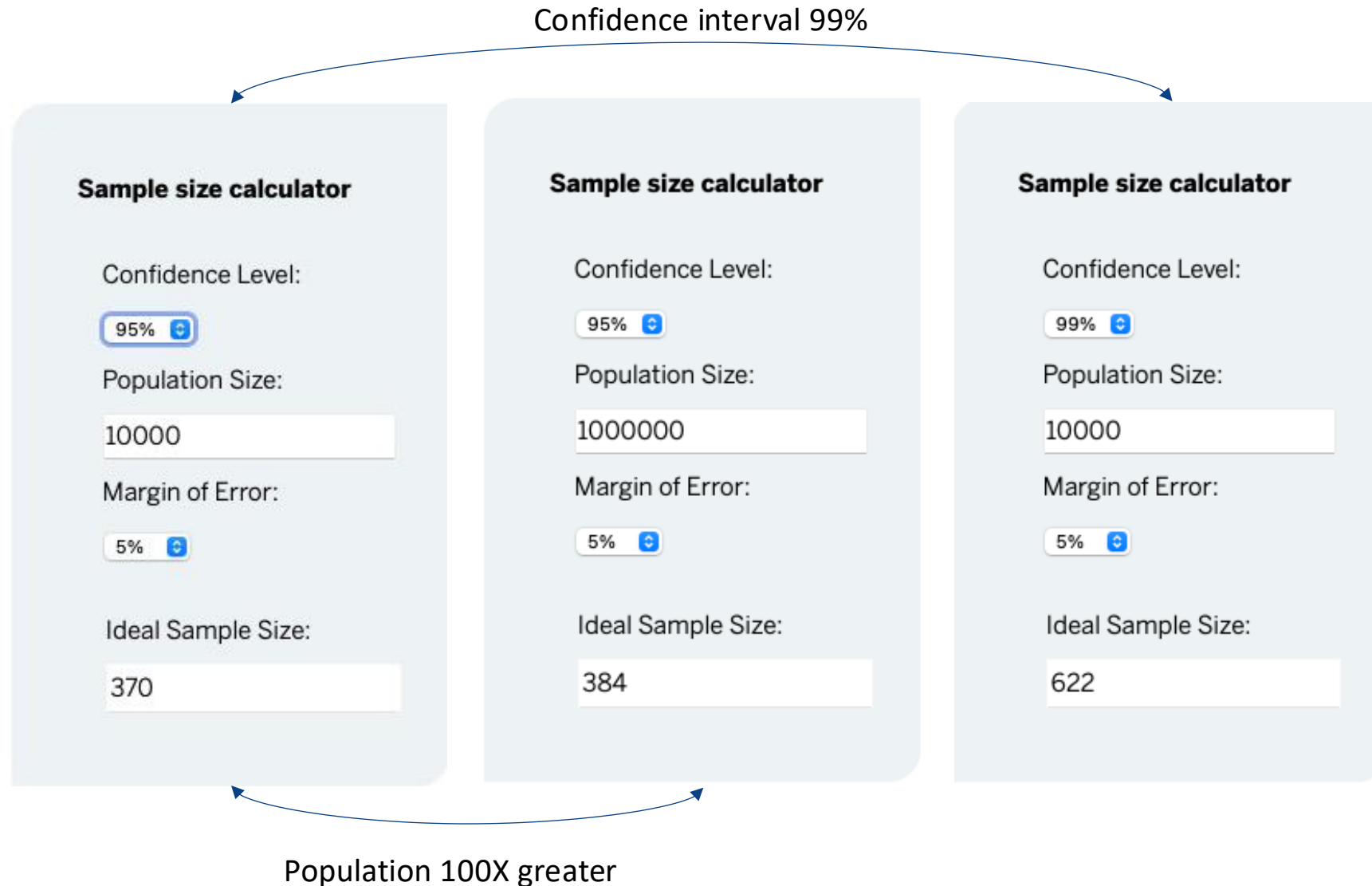
$$E = z \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \quad \longrightarrow \quad n = \hat{p}(1 - \hat{p}) \left(\frac{z}{E}\right)^2$$

If prior research shows that 20% of people surveyed have never reviewed a restaurant, what is the size of the sample to construct a 95% confidence interval with a margin of error of 5%?

$$n = 20\% * (1-20\%) * (1.96/5\%)^2 = 245.9 \rightarrow 246$$

Qualtrics Sample Size Calculator

<https://www.qualtrics.com/blog/calculating-sample-size/>





How to determine nonprobability sample sizes?

The sample size formula cannot be used for nonprobability samples

- Determining the sample size is a subjective, intuitive judgment

Sample sizes are often determined using less formal approaches



Avoiding Nonresponse

Lecture 5, part 3

Nonresponse

Nonresponse characterizes surveys and polls. Few participate.

This has led researchers to conclude we are surveying “the tails of the distribution” – we are not obtaining representative samples.

What to do?

1. Design choices (in W5)

- Design good questions
- Design good surveys
- Consider administration
- Pretest

2. Analysis choices

- Missing data treatment



Design Good Surveys

Based on Don Dillman!

1. pre-notice letter-establish a bond
2. send out survey and sponsor letter
3. thank you/reminder
4. replacement questionnaire
5. special procedures (calling & other channels)

